AI Market Twin — Detailed Functional Specification (Developer Handoff)

---

1. Document Purpose

This document defines detailed product requirements, system behavior, user flows, data entities, APIs, permissions, and non-functional requirements for engineering implementation.

Audience:

- Frontend Engineers
- Backend Engineers
- AI/ML Engineers
- QA Engineers
- Product Managers
- Designers

---

2. Product Summary

AI Market Twin is a B2B SaaS platform that simulates product launch outcomes using synthetic AI personas and predictive scoring.

Core value:

- Predict success probability
- Recommend launch country
- Optimize pricing
- Identify target audience
- Compare creative performance
- Reduce launch risk

---

3. User Roles

3.1 Owner

- Full workspace access
- Billing admin
- Team management
- All reports

3.2 Admin

- Manage projects
- Run simulations
- Export reports
- Invite members

3.3 Analyst

- Create/edit projects
- Run simulations
- View results

3.4 Viewer

- Read-only dashboards/reports

---

4. Core Modules

1. Authentication
2. Workspace Management
3. Project Management
4. Simulation Engine
5. Results Dashboard
6. Reporting & Export
7. Billing & Subscription
8. Notifications
9. Admin Console
10. Analytics & Logs

---

5. Functional Requirements

Module A — Authentication

A1 Login

Methods:

- Email + Password
- Google OAuth
- SSO (Enterprise future)

Validation:

- Email format required
- Password min 8 chars

Features:

- Forgot password
- Email verification
- Session management
- Logout all devices

APIs

POST /auth/register POST /auth/login POST /auth/logout POST /auth/forgot-password POST /auth/reset-password GET /auth/me

---

Module B — Workspace Management

B1 Workspace Create

Fields:

- Workspace name
- Company name
- Industry
- Country

B2 Team Invite

Fields:

- Email
- Role

B3 Role Permissions

RBAC enforced on API and UI.

APIs

POST /workspaces GET /workspaces/:id POST /workspaces/:id/invite PATCH /workspaces/:id/member/:userId DELETE /workspaces/:id/member/:userId

---

Module C — Project Management

C1 Create Project

Fields:

- Project name
- Product name
- Category
- Description
- Base price
- Currency
- Launch objective
- Candidate countries[]
- Competitor URLs[]
- Creative assets[]

Status:

- Draft
- Ready
- Running
- Completed
- Failed
- Archived

C2 Edit Project

Editable until simulation starts.

C3 Duplicate Project

Clone settings for A/B testing.

APIs

POST /projects GET /projects GET /projects/:id PATCH /projects/:id DELETE /projects/:id POST /projects/:id/duplicate

---

Module D — File Uploads

Supported:

- JPG
- PNG
- MP4
- PDF (future)

Constraints:

- Max 200MB per file
- Virus scan required
- Signed URL uploads

Storage:

- S3 / object storage

APIs

POST /uploads/presign POST /uploads/complete DELETE /uploads/:id

---

Module E — Simulation Engine

E1 Trigger Simulation

Input:

- projectId
- modelVersion
- simulationSize (10k default)

Process Stages: 1. Validate input 2. Generate personas 3. Score countries 4. Run pricing model 5. Evaluate creatives 6. Calculate risk 7. Generate recommendations 8. Persist results

E2 Job Queue

Use async workers. Recommended:

- Celery / BullMQ / Temporal

E3 Retry Logic

- Max 3 retries
- Dead-letter queue

APIs

POST /simulations/:projectId/run GET /simulations/:id/status GET /simulations/:id/logs POST /simulations/:id/cancel

---

Module F — Persona Engine

Persona Schema

- personaId
- ageRange
- gender
- country
- incomeBand
- profession
- interests[]
- purchaseStyle
- priceSensitivity
- trustFactors[]
- objections[]

Output Size

Default 10,000 records (stored aggregated unless enterprise raw access)

Module G — Scoring Models

Metrics

- Success Score (0–100)
- Purchase Intent Score
- CTR Prediction
- Conversion Probability
- Failure Risk
- Competition Pressure
- Recommended Price
- Country Rank Score

Versioning

All scoring models versioned. Store version in results.

Module H — Results Dashboard

H1 Overview Tab

Cards:

- Success Probability
- Best Country
- Best Audience
- Best Price
- Best Creative
- Risk Level

H2 Countries Tab

Table Columns:

- Country

- Demand Score
- CAC Estimate
- Competition Score
- Final Score
- Priority Rank

H3 Personas Tab

List/Grid with filters:

- Age n- Country
- Intent level
- Income

H4 Pricing Tab

Charts:

- Price vs Conversion
- Price vs Revenue
- Margin estimate

H5 Creative Tab

Creative asset scorecards.

H6 Recommendation Tab

Narrative strategy output.

APIs

GET /results/:simulationId/overview GET /results/:simulationId/countries GET /results/:simulationId/personas GET /results/:simulationId/pricing GET /results/:simulationId/creative GET /results/:simulationId/recommendations

---

Module I — Reporting

Export Formats

- PDF

- PPTX (future)
- CSV (tables)
- Share link

PDF Sections

1. Executive Summary
2. Key Metrics
3. Country Ranking
4. Pricing Recommendation
5. Persona Insights
6. Risks
7. Action Plan

APIs

POST /reports/:simulationId/pdf POST /reports/:simulationId/share-link GET /reports/:id/download

---

Module J — Billing

Plans

Starter / Growth / Enterprise

Features by Plan

- Monthly simulation quota
- Team seats
- Export limits
- Raw data access
- API access

Payment

- Stripe recommended
- Invoice for Enterprise

APIs

GET /billing/plans POST /billing/checkout GET /billing/subscription POST /billing/webhook

---

Module K — Notifications

Channels:

- In-app
- Email
- Slack (future)

Events:

- Simulation completed
- Export ready
- Invite accepted
- Quota exceeded
- Payment failed

Module L — Admin Console

Internal only. Features:

- User management
- Workspace suspension
- Usage analytics
- Job monitoring
- Model version rollout
- Support impersonation (logged)

---

6. Database Entities

Core Tables

- users
- workspaces
- workspace_members
- projects
- project_assets
- simulations
- simulation_results

- reports
- subscriptions
- invoices
- notifications
- audit_logs

---

7. Suggested Tech Stack

Frontend

- React / Next.js
- TypeScript
- Tailwind
- Recharts

Backend

- Python FastAPI or Node NestJS

AI Layer

- LLM APIs
- Python pipelines
- Vector DB optional

Infra

- AWS / GCP
- PostgreSQL
- Redis
- Queue workers

---

8. Non-Functional Requirements

Performance

- Dashboard load < 2 sec cached
- Simulation status real-time updates
- 10k persona job < 5 min target

Security

- JWT / secure sessions
- RBAC
- Encryption at rest
- Signed file URLs
- Audit logs
- OWASP controls

Reliability

- 99.9% uptime target
- Daily backups
- Retry jobs

Scalability

- Horizontal worker scaling
- Multi-tenant architecture

---

9. QA Acceptance Criteria

Project Create

Given valid input, project saved successfully.

Simulation Run

Given ready project, job enters queue and completes with results.

Export PDF

Given completed simulation, PDF downloadable.

RBAC

Viewer cannot edit project.

Billing

Quota enforced after plan limit reached.

---

10. Milestone Release Plan

v0.1 MVP

- Auth
- Projects
- Run simulation
- Overview dashboard
- PDF export

v0.2 Growth

- Team roles
- Pricing optimizer
- Creative tests
- Billing

v1.0 Enterprise

- API
- SSO
- White label
- Advanced analytics

---

11. Open Questions

- Raw persona storage policy?
- Which model providers initially?
- Regulatory disclaimers by region?
- Human review layer needed?
- Data retention policy by plan?

---

12. Final Delivery Note

Engineering should prioritize speed to MVP, observability, and monetization-ready architecture.